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A review: Knowledge reasoning over knowledge graph

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Abstract

Mining valuable hidden knowledge from large-scale data relies on the support of reasoning technology. Knowledge graphs, as a new type of knowledge representation, have gained much attention in natural language processing. Knowledge graphs can effectively organize and represent knowledge so that it can be efficiently utilized in advanced applications. Recently, reasoning over knowledge graphs has become a hot research topic, since it can obtain new knowledge and conclusions from existing data. Herein we review the basic concept and definitions of knowledge reasoning and the methods for reasoning over knowledge graphs. Specifically, we dissect the reasoning methods into three categories: rule-based reasoning, distributed representation-based reasoning and neural network-based reasoning. We also review the related applications of knowledge graph reasoning, such as knowledge graph completion, question answering, and recommender systems. Finally, we discuss the remaining challenges and research opportunities for knowledge graph reasoning.

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Multi-hop reasoning on Knowledge Graphs (KGs) aims at inferring the triplets that not in the KGs to address the KGs incompleteness problem. Reinforcement learning (RL) methods, exploiting an agent that takes incremental steps by sampling a relation and entity (called an action) to extend it...

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